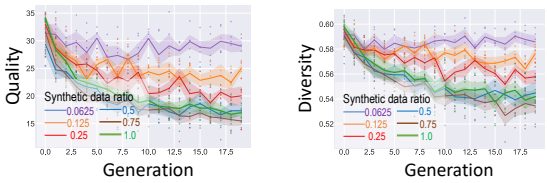

Figure 2: **Evolution of quality (left) and diversity (right) over generations for different synthetic data ratios on the *100M_tweets* dataset.** Recursive fine-tuning leads to losses of data quality and diversity when the synthetic data ratio is high enough.

The Semantic Diversity of a set of texts is measured as the pairwise cosine diversity in the `stella_en_1.5B_v5` model (Zhang et al., 2024a), as in previous studies (Guo et al., 2024). **Quality** and **Political lean** are estimated by using `LLaMa-3.3-70B-Instruct` (Dubey et al., 2024) in the LLM-as-a-Judge setup. The prompt is inspired by Wang et al. (2023) and Chen et al. (2023), and adapted to our task of evaluating the quality of short texts (see appendix B.3 or details and validation of that adaptation). In sections 4.3 and 4.4, we additionally rely on the following metrics. **Lexical Diversity** is estimated as SelfBLEU (Zhu et al., 2018), computed as the average BLEU score (Papineni et al., 2002) of each text using all other texts as references, following prior work (Guo et al., 2024). **Gaussianity** is measured by fitting a 2D UMAP projection on embeddings from the `stella_en_1.5B_v5` model, and computing the AIC (Akaike, 1974) of a 2D Gaussian distribution fit to this space. **Positivity** is assessed using the `SentimentIntensityAnalyzer` tool from NLTK (Hardeniya et al., 2016), which assigns a sentiment score to each text ranging from -1.0 (highly negative) to 1.0 (highly positive).

4 Experiments

In this section, we study the following questions:

- Does synthetic data ratio impact distribution shift dynamics?
- Do different datasets exhibit different distribution shifts dynamics?
- Which dataset properties are associated with distribution shift dynamics?
- Does training on multiple domains influence distribution shift dynamics?

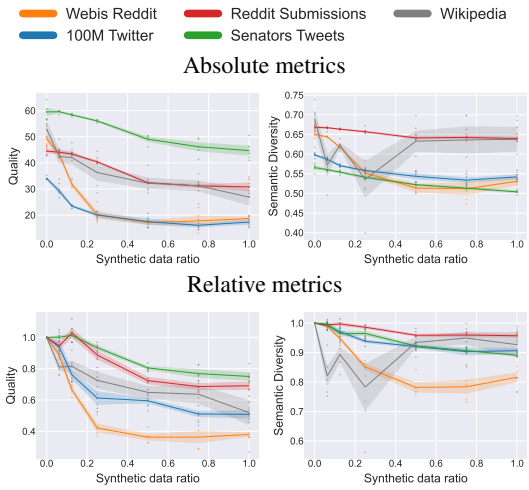

Figure 3: **Effect of synthetic data ratio on absolute and relative quality (left column) and diversity (right column) at the last generation, in four different datasets.** Absolute measures (top row) correspond to the value of the corresponding metric at generation 19. Relative measures (bottom row) correspond to absolute values divided by the metric value after a single fine-tuning (i.e. generation 0). Different datasets lead to different sensitivities to synthetic data ratio, with *100M_tweets* (blue) and *webis_reddit* (orange) exhibiting greater losses in quality and diversity.

- Do datasets with different political biases lead to different shifts in political lean?

4.1 Does synthetic data ratio impact distribution shift dynamics?

In this experiment, we aim to reproduce the effect of the synthetic data ratio (Briesch et al., 2023) on the shifts toward lower quality and lower diversity induced by recursive fine-tuning. This ratio corresponds to the proportion between the AI-generated data and the freshly sampled human data that are added to the Accumulated data pool at each generation.

Figure 2 shows the evolution of quality and semantic diversity over consecutive generations, where human data comes from the *100M_tweets* dataset. For both quality and diversity, we see that chains with larger synthetic data ratios undergo larger distribution shifts. Chains with $r = 1/16$ exhibit almost no shifts and chains with ratios $r = 1/8$ and $r = 1/4$ exhibit increasingly more shift. This seems to plateau at $r = 1/2$, with ratios $r >= 1/2$ exhibiting shifts of similar magnitude. This experiment shows that, for the *100M_tweets* dataset, synthetic data ratio has a significant impact on distribution shift dynamics, and that the loss in

quality and diversity appears to plateau when half or more training data is synthetic.

4.2 Do different datasets exhibit different distribution shifts dynamics?

In this experiment, we explore how the distribution shift dynamics vary over different datasets. The experiment is methodologically identical to the one in 4.1, but we consider five datasets: *100M_tweets* , *senator_tweets* , *reddit_submissions* , *webis_reddit* , and *wikipedia* .

Figure 3 shows the values of the metrics at the end of the iterative chain (i.e. those measured at generation 19 in Figure 2) as a function of synthetic data ratio. Figure 3 shows both absolute (top) and relative (bottom) quality (left) and diversity (right) scores. *Relative scores* correspond to absolute scores divided by the score of the data generated in generation zero. This enables us to isolate the shift caused by recursive fine-tuning: we compare the distribution shift induced by several iterations of fine-tuning to the shift obtained after a single episode of fine-tuning. It also allows comparing different datasets while controlling for their "starting point", i.e. the value of quality and diversity in the human dataset. Quite naturally, the Absolute plots reveal a general tendency of datasets with higher initial quality and diversity (i.e. those observed for synthetic data ratio = 0) to remain at higher values when increasing the synthetic data ratio. Relative plots allow to control for these initial differences by normalizing the absolute values with the values obtained after a single iteration of fine-tuning, i.e. *relative loss* in quality and diversity. *webis_reddit* exhibits a relative loss both in quality and diversity. *reddit_submissions* and *senator_tweets* datasets exhibit small relative losses in quality and diversity, and *100M_tweets* dataset also exhibits a small relative loss in diversity. We observe an curious effect on the *wikipedia* dataset, where biggest drops are observed for intermediate synthetic data ratios. Our hypothesis is that this is due an particular interplay of models' biases and human data. In appendix C.3, we discuss this hypothesis in more detail and conduct a toy experiment to provide further support for this hypothesis. Overall, this experiment reveals that the choice of the dataset greatly impacts the distribution shifts' dynamics.

4.3 Which dataset properties are associated with distribution shift dynamics?

The previous sections indicate that the extent to which recursive fine-tuning leads to distribution shifts varies greatly between datasets. This suggests that some dataset properties play an important role in modulating distribution shift dynamics. In this section, we describe a series of regression analysis experiments aimed at uncovering which properties have such strong influence on distribution shifts. We focused on six dataset properties as relevant candidates: Semantic diversity (using pair-wise cosine diversity), Lexical diversity (using self-BLEU), Gaussianity, Quality, Positivity and Text length (see Appendix B.3.1 for details on how those were selected). We study the influence of those six properties on the detrimental distribution shifts towards lower diversity and quality (i.e. model collapse).

We extracted 200 clusters from four of the five datasets from the previous section, using the method described in appendix B.4 (the *senator_tweets* dataset was excluded due to its insufficient size). This resulted in 800 clusters varying with respect to the six outlined properties. We used those properties as predictors in our regression analysis. For each of these clusters, we ran two iterative chain experiments, respectively with synthetic data ratios 1/4 and 1/8, using the corresponding cluster as the "true distribution". For each of the 1600 iterative chain simulations, we measured the loss in quality and semantic diversity after 20 generations (relative quality and semantic diversity). We used those values as dependent variables. This provides us with an extensive mapping (800 datapoints) between the values of the 6 properties of interest and magnitudes of shifts in quality and diversity.

By performing regression analyses, we were then able to determine which properties correlate with distribution shifts. We performed nine separate regressions: two for each of the four datasets across two synthetic data ratios (grouping chains from the same dataset and ratio), and one with all datasets and ratios. Table 1 show the results, with columns corresponding to different regression analyses. Statistically significant coefficients ($p < 0.05$) are shown in bold. Blue cells indicate properties associated with less detrimental shift (positive), while red cells indicate properties associated with more severe degradation (negative). Regression analyses conducted on all data outlined

Coefficient	All	*webis_reddit*		*100M_tweets*		*reddit_submissions*		*wikipedia*	
Synthetic data ratio		1/8	1/4	1/8	1/4	1/8	1/4	1/8	1/4
Semantic Diversity									
Semantic diversity	-0.0007	**0.0075**	0.0117	0.0087	**0.0291**	0.0003	0.0057	0.0152	0.0181
Lexical diversity	**-0.0126**	**-0.0155**	**-0.0487**	-0.0012	**-0.0051**	-0.0026	**-0.0098**	**-0.0306**	-0.0091
Gaussianity	**-0.0092**	-0.0001	-0.0006	-0.0042	**-0.0121**	0.0016	-0.0025	**-0.0325**	-0.0170
Quality	**0.0187**	-0.0003	-0.0021	-0.0006	**0.0105**	0.0027	0.0040	**0.0410**	0.0283
Positivity	-0.0031	**-0.0037**	**-0.0071**	0.0007	0.0004	-0.0050	**-0.0153**	-0.0050	**0.0094**
Text length	-0.0015	0.0005	0.0114	-0.0015	**-0.0214**	-0.0049	**-0.0154**	-0.0356	**-0.0407**
Quality									
Semantic diversity	**0.0105**	**0.0410**	**0.0185**	0.0140	0.0316	0.0129	0.0058	-0.0177	0.0219
Lexical diversity	**-0.0603**	**-0.0892**	**-0.0478**	-0.0043	**-0.0275**	0.0080	0.0107	-0.0240	**-0.0719**
Gaussianity	**-0.0158**	0.0012	-0.0095	-0.0102	-0.0139	-0.0026	0.0033	-0.0159	-0.0122
Quality	**0.0616**	**0.0335**	-0.0018	-0.0044	**0.0547**	0.0113	0.0055	0.0297	**0.1023**
Positivity	**0.0074**	-0.0055	**-0.0097**	**0.0304**	**0.0380**	0.0070	**-0.0431**	-0.0073	0.0106
Text length	**-0.1327**	**-0.1029**	0.0037	0.0157	**-0.0554**	-0.0244	-0.0181	-0.0414	**-0.1218**
				$p < 0.05$	$p < 0.01$	$p < 0.001$	$p < 0.05$	$p < 0.01$	$p < 0.001$

Table 1: **Regression coefficients for distribution shifts in semantic diversity and quality.** Bold values indicate statistical significance. Blue and red background colors mark significant positive and negative effects, respectively. Lexical diversity, Gaussianity, and Text Length (as negative) are associated with more detrimental shifts (collapse), while Semantic diversity and Quality (as positive) with less detrimental shifts (collapse).

the following properties as significant. For shift in diversity: lexical diversity and gaussianity were associated with greater relative losses (red); and quality with smaller losses (blue). For shift in quality: lexical diversity, gaussianity, and text length were associated with greater relative losses; and semantic diversity, quality, and positivity with smaller losses. Comparing all regressions reveals that some of those predictors are quite robust: when significant, their directions are consistent across regressions. Lexical diversity is the most robust predictor appearing 11 times. Quality and length appear 7 times and often together. Semantic diversity appears 5 times and Gaussianity 4 times. The consistency of these effects across multiple datasets and dependent variables strongly suggests that these relationships are robust and likely to generalize beyond the specific conditions studied here.

4.4 What happens when models are trained on data from multiple domains?

So far we have considered situations in which models are trained on data from a single Internet domain. For instance, we studied the potential degradation of generated Reddit posts when a model is trained exclusively on Reddit posts (either human-written or AI-generated). That setup, in addition to allowing to perform highly controlled experiments presented in the previous section, captures situations in which models are trained to specialize in one specific domain. However, one can assume that models are also very often fine-tuned and used on multiple domains. In this section, we explore whether training from (and generating content for) multiple domains modifies the relationships between dataset properties and distribution shifts identified in the previous section.

On the one hand, we could expect models trained on data from multiple domains to merge internal representations, and therefore that the properties of content from a given domain (e.g. Reddit) influences the content generated in another domain (e.g. Wikipedia) (*Hypothesis 1*). Alternatively, models may keep the representations of different domains separate, in which case the properties of data in one domain would not influence on the generation in another domain (*Hypothesis 2*). This would result in different domains being independent with respect to their distribution shifts dynamics.

To explore that question, we slightly modified the experimental pipeline from the previous sections. Having already extracted 200 clusters from *wikipedia* , *webis_reddit* and *100M_tweets* datasets, we merge those 600 "pure" clusters into 200 "mixed" clusters so that each "mixed" cluster consists of one "pure" cluster from each dataset (i.e. $mixed_i = wikipedia_i \cup webis_reddit_i \cup 100M_tweets_i, i \in [1, 200]$). We then run 200 iterative chain experiments using those clusters. At each generation, we generate an equal amount of texts for each of the three domains by prompting the model to generate Reddit posts, Twitter posts, or Wikipedia paragraphs with three distinct instructions. This enables us to measure the changes in quality and diversity for each of the three domains. We map those changes to the properties of